Exploring eDNA in SBDI

The **S**wedish **B**iodiversity **D**ata **I**nfrastructure - a national access point for **open biodiversity data**

We provide **tools and services** for interdisciplinary research on biodiversity and ecosystems

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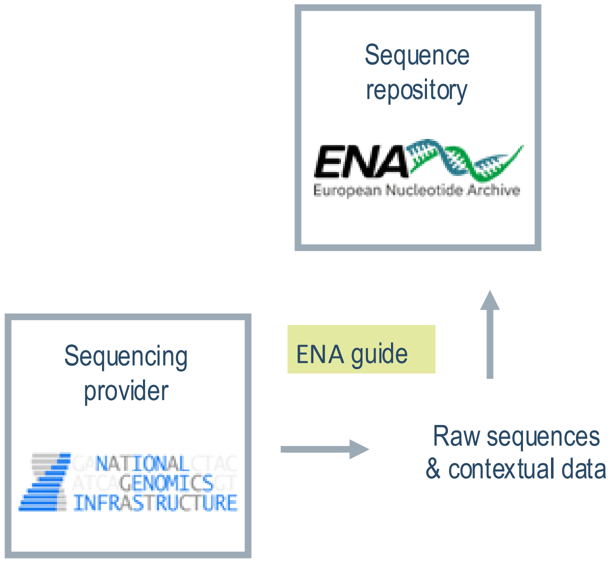
A **guide** for simplified submission of raw reads and metadata to **ENA**



Step 3: Register samples

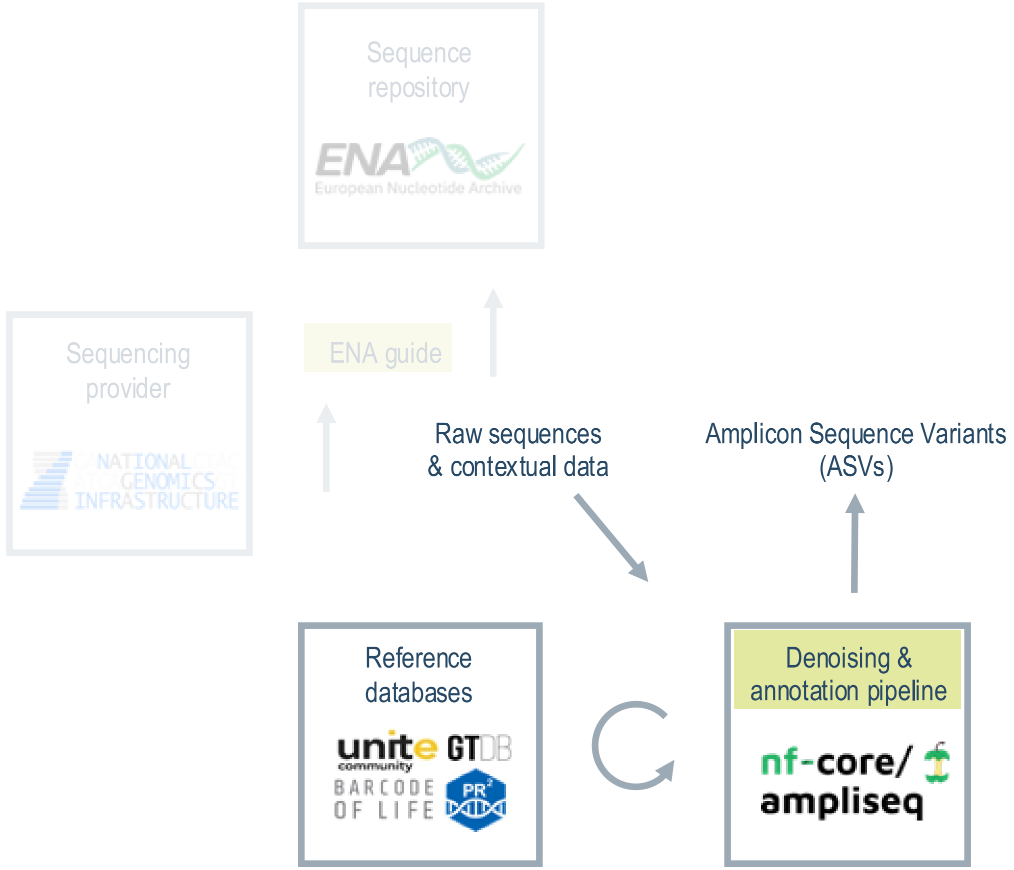
Samples are the source material from which your sequences derive, and the searchability and usability of your submitted data will depend on how well you document these samples. Go to *Samples* | *Register Samples* and click *Download spreadsheet to register samples* to start the process.

Step 3a: Select sample checklist

The ENA sample checklists are partly overlapping sets of attributes (or data fields) that can be used to describe samples, and by selecting one of these you enable your sample metadata to be validated for correctness during submission. For environmental and organismal (host-associated) samples, alike, we recommend using one of the *Environmental Checklists* and, among these, to select the alternative from the *Genomic Standards Consortium (GSC) MIxS checklists* that provides the most specific match to your sampled environment, for example:

Sampled environment	Recommended checklist
Air or general, above-ground, terrestrial	GSC MixS air
Epi- or endophytic (e.g. leaf, root)	GSC MIxS plant associated
Epi- or endozoic (e.g. spider gut, animal skin)	GSC MIxS host associated
Fresh- or seawater	GSC MixS water
Human gut / oral / skin / vaginal	GSC MIxS human gut / oral / skin / vaginal
Human non- gut / oral / skin / vaginal	GSC MIxS human associated
Sediment	GSC MixS sediment
Soil	GSC MixS soil

A **pipeline for denoising** reads into Amplicon Sequence Variants (**ASVs**)



nf-co.re/ampliseq/2.18.0

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nf-core/ampliseq

Amplicon sequencing analysis workflow using DADA2 and QIIME2

16s 18s amplicon-sequencing edna illumina iontorrent its metabarcoding metagenomics metataxonomics microbiome pacbio qiime2 rrna taxonomic-classification taxonomic-profiling

Launch version 2.18.0

https://github.com/nf-core/ampliseq

Daniel Lundin (LnU)

Introduction | Usage | Parameters | Output | Results | Releases | 2.18.0

Introduction

nfcore/ampliseq is a bioinformatics analysis pipeline used for amplicon sequencing, supporting denoising of any amplicon and supports a variety of taxonomic databases for taxonomic assignment including 16S, ITS, CO1 and 18S. Phylogenetic placement is also possible. Multiple region analysis such as 5R is implemented. Supported is paired-end Illumina or single-end Illumina, PacBio and IonTorrent data. Default is the analysis of 16S rRNA gene amplicons sequenced paired-end with Illumina.

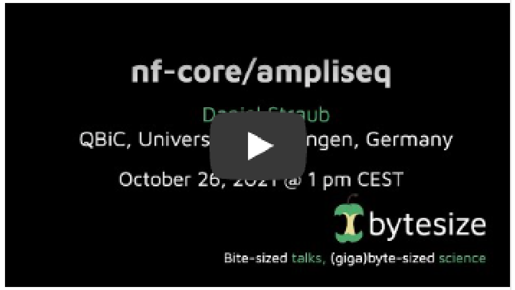
A video about relevance, usage and output of the pipeline (version 2.1.0; 26th Oct. 2021) can also be found in YouTube and billibilli, the slides are deposited at figshare.

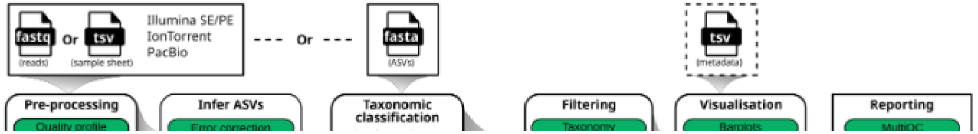
run with

```
nf-core pipelines launch nf-co
```

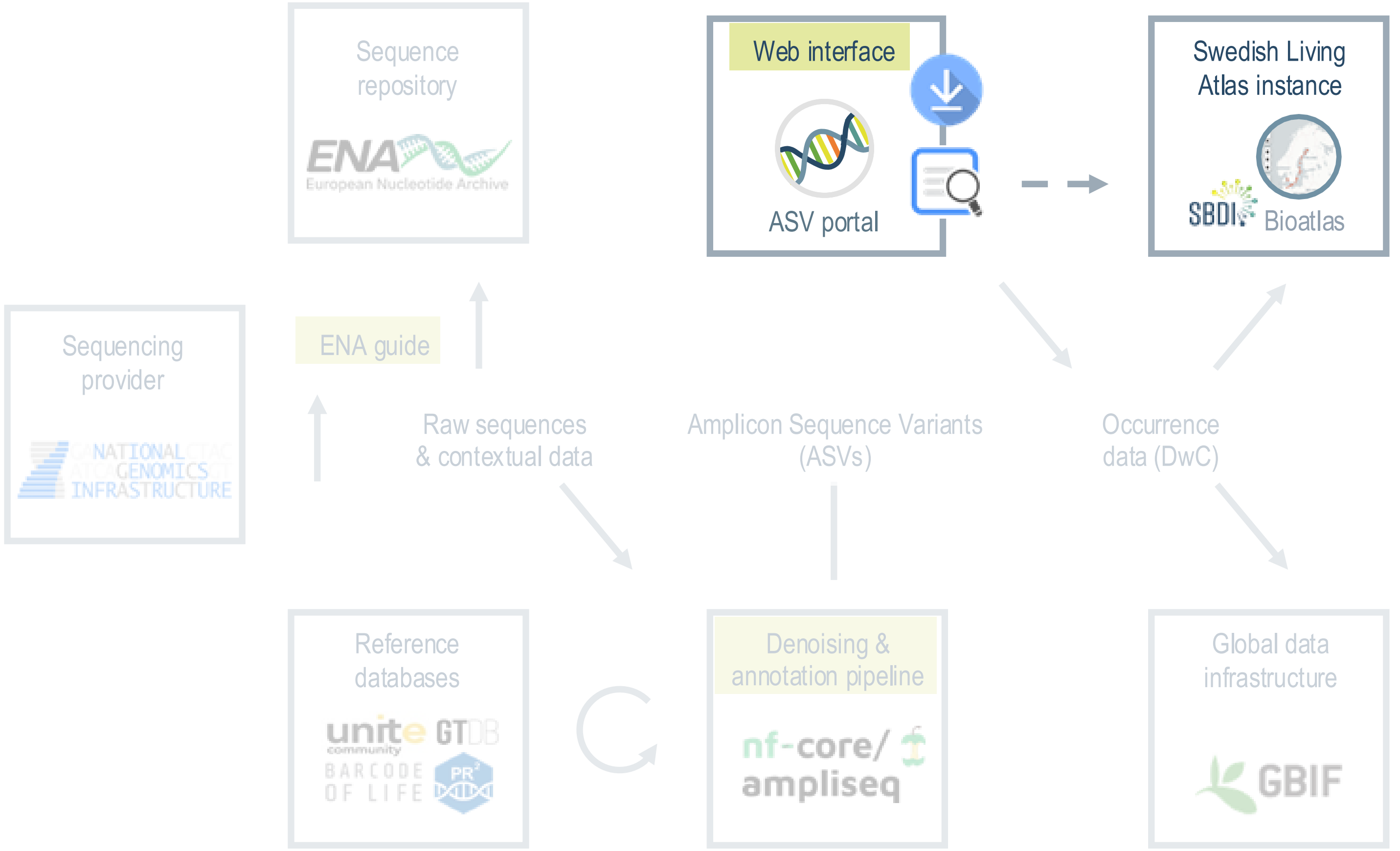
nf-core | Nextflow | Seqera Platform

video introduction





A **web interface** for submitting and accessing ASV data in SBDI



asv-portal.biodiversitydata.se/index

SBDI Swedish Biodiversity Data Infrastructure

All SBDI tools
Support form

SWEDISH ASV PORTAL | BLAST SEARCH | FILTER SEARCH | SUBMIT DATA | DOWNLOAD DATA | ABOUT | LOGIN

Swedish ASV portal

Explore and share DNA sequences and species occurrences derived from metabarcoding (eDNA) studies

Search for ASVs and Bioatlas records using Basic Local Alignment Search Tool (BLAST)

BLAST

Search for ASVs and Bioatlas records using filters on sequencing details and taxonomy

FILTER

Submit your metabarcoding dataset to the ASV database and SBDI Bioatlas

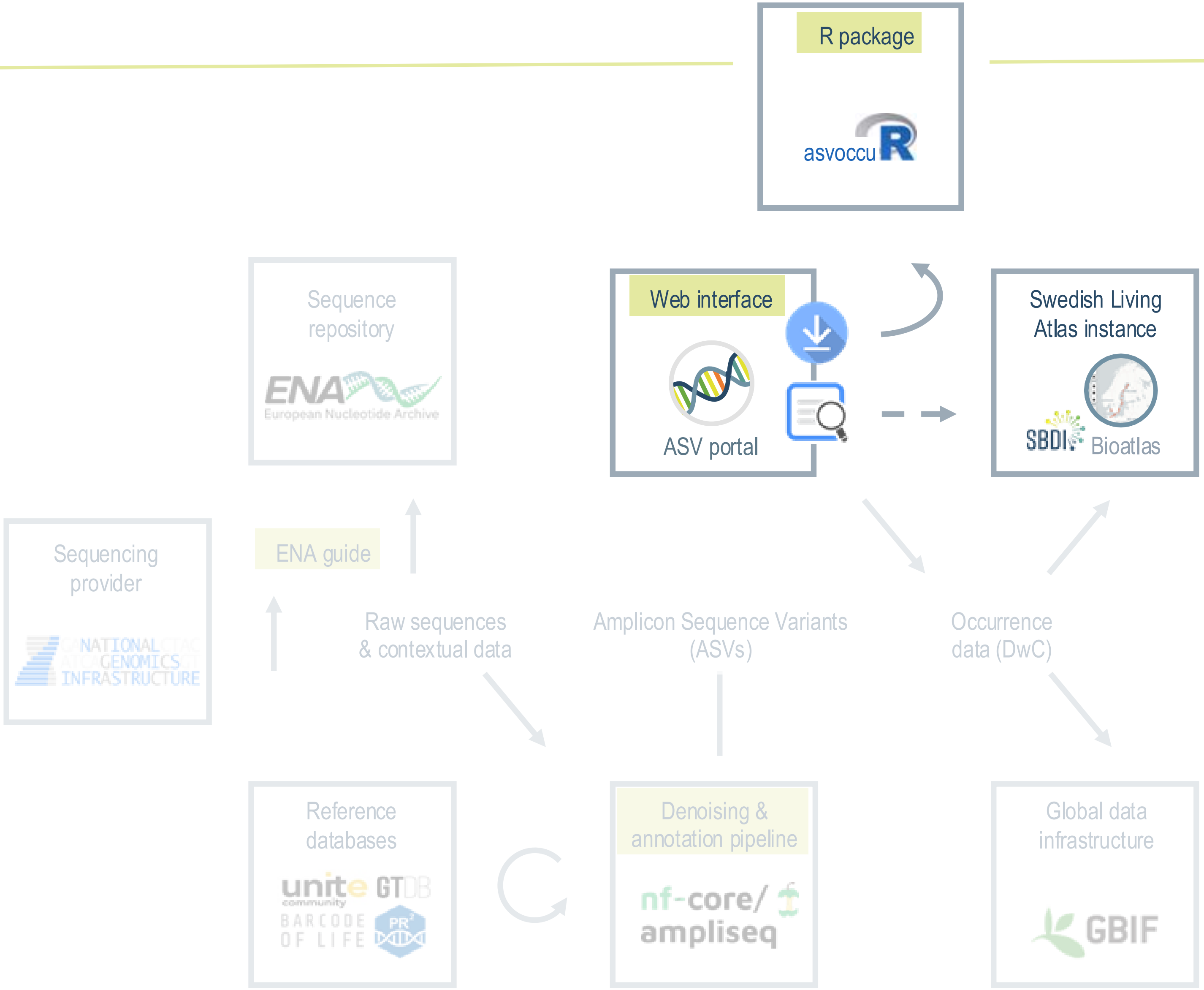
SUBMIT

Download ASV occurrence datasets, in a condensed Darwin Core-like format

DOWNLOAD

Accessibility
Privacy Policy

An **R package** for processing downloaded data into ASV table format



github.com/biodiversitydata-se/asvoccur

asvoccur.Rproj Merge branch 'release/1.0.0' last week

README License MIT license

asvoccur

R tools for ASV occurrence data in SBDI.

Overview

The **asvoccur** R package, currently under development, provides tools for unpacking and processing ASV occurrence data and metadata downloaded from the Swedish ASV portal. It enables users to convert condensed DwC archives into ASV table format for easier downstream analysis in R, by using functions that load, merge, and aggregate ASV counts across taxonomic ranks.

Install

```
install.packages('remotes')
remotes::install_github("biodiversitydata-se/asvoccur")
# or:
# remotes::install_github("biodiversitydata-se/asvoccur@develop")
library(asvoccur)
```

GitHub installations (local source builds) may occasionally fail. If so:

```
remove.packages("asvoccur")
# restart R / RStudio
remotes::install_github("biodiversitydata-se/asvoccur", force = TRUE)
```

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Today's tutorial

ENA submission guide

Pipeline: nf-core/ampliseq

Web interface: ASV portal

R-package: asvoccur